How varied are stellar populations in the Solar vicinity? Exploring GCNS and GALAH DR4 with t-SNE

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Abstract

We select a set of stars in the Solar vicinity observed by the Gaia Catalogue of Nearby Stars (GCNS; which comprises approximately 330k stars within 100 pc of the Sun) [2] that have also been followed-up with high-resolution spectroscopy by the GALactic Archaeology with HERMES (GALAH) DR4 survey [1]. In this context, the visualization of multidimensional data with potentially non-linear relationships remains a significant challenge. Our goal is to verify whether t-distributed Stochastic Neighbor Embedding (t-SNE) can retrieve the distinct stellar population properties in an unsupervised manner (effectively by clustering similar stars together), and to explore whether peculiar stars, such as emission-line objects, stand apart in the map (outlier detection itself is a later step). Hence, our study focuses on exploring how the t-SNE captures the main spectral features of the ∼5,000 stars in our sample. We tested and validated the t-SNE projection through spectral parameter-based color-mapping, which revealed a smooth gradient across the projected data, demonstrating that the algorithm successfully preserves the intrinsic physical correlations between several stellar parameters, such as $T_{\rm eff}$, $\log g$, and [Fe/H]. Furthermore, we examined the perplexity hyperparameter as a weighting factor between manifold scales, noting that a value of 30 increases sensitivity to local structures, whereas a value of 50 prioritizes the preservation of the global topology. Our preliminary results indicate the presence of a small detached sub-cluster, which maintains a different internal parameter gradient, distinguishing it from the main stellar group as a unique population within the local manifold, which requires further investigation.

1. Introduction

The cross-match between GCNS [2] and GALAH DR4 [1] within 100 pc was characterised at catalogue level in our SAB25 proceedings [7]:

∼6,000	FGK	1.5 Gyr	−0.19	Thick disk
stars in common	dwarf dominated	median age	median [Fe/H] (dex)	plus probable halo

High-resolution spectra are high-dimensional and carry subtle, non-linear variations that linear projections blur. Manifold learning embeds them in two dimensions directly from flux morphology, with no synthetic grids or labels; the methods are defined in Section 2.

2. Data and Methodology

Step	DATA	MAP	FILTER	GROUP	CHECK	What happens
1 Sample	■					GCNS × GALAH DR4: **6 198** stars within 100 pc [1,2]; FGK dwarfs dominate.
2 Spectra	■					4 CCDs, native pixels, **14 245 px** per star; 894 lack a full set, so **5 304** are mapped.
3 Manifold		■				*t*-SNE (perplexity 30, 50) and UMAP (15, 30 neighbours); cosine metric, no PCA, no z-score.
4 Cuts & flags			■			Display filters $T_{\rm eff} \geq 4500$ K and S/N ≥ 10 in 4 CCDs; flags: emission, SB1, SB2, low S/N; no re-projection.
5 Colouring			■			$T_{\rm eff}$, $\log g$, [Fe/H] only colour the map, never as labels: do pixels alone group stars by atmosphere?
6 Emission				■		HDBSCAN [6] on the 286 emission-flagged stars: dense groups of any shape, subclusters S0–S4.
7 Evaluation					■	15-NN overlap (0.21–0.37), distance correlation (0.38–0.49), $R^2 \approx 0.9$ for $T_{\rm eff}$.
8 Validation					■	Emitters vs. quiet dwarfs, line by line: Hα (6563 Å) and TiO bands.

PCA vs. *t*-SNE & UMAP
PCA is linear and blurs subtle differences (not applied); *t*-SNE [3,4] and UMAP [5] are non-linear and keep similar spectra together.

Perplexity
Neighbours each star "looks at": 30 favours local structure, 50 the global topology.

Overlap & distance corr.
Overlap: share of the 15 nearest neighbours kept in 2D (local). Distance correlation: Spearman of pairwise distances (global).

HDBSCAN
Dense groups of any shape, no preset number; stars outside any dense group stay unassigned.

3. Results

Figure 1 — Main unsupervised projection (*t*-SNE, perplexity 30; 5 304 stars). **(a)** Coloured by effective temperature ($T_{\rm eff}$); markers: emission (Hα), SB1/SB2 (binary candidates) and low S/N. One point is one star; close points are similar spectra; axes have no units. **(b)** Same map coloured by $T_{\rm eff}$, $\log g$ and [Fe/H]. These properties never enter the algorithm; they only test whether spectral similarity alone groups stars by atmospheric properties.

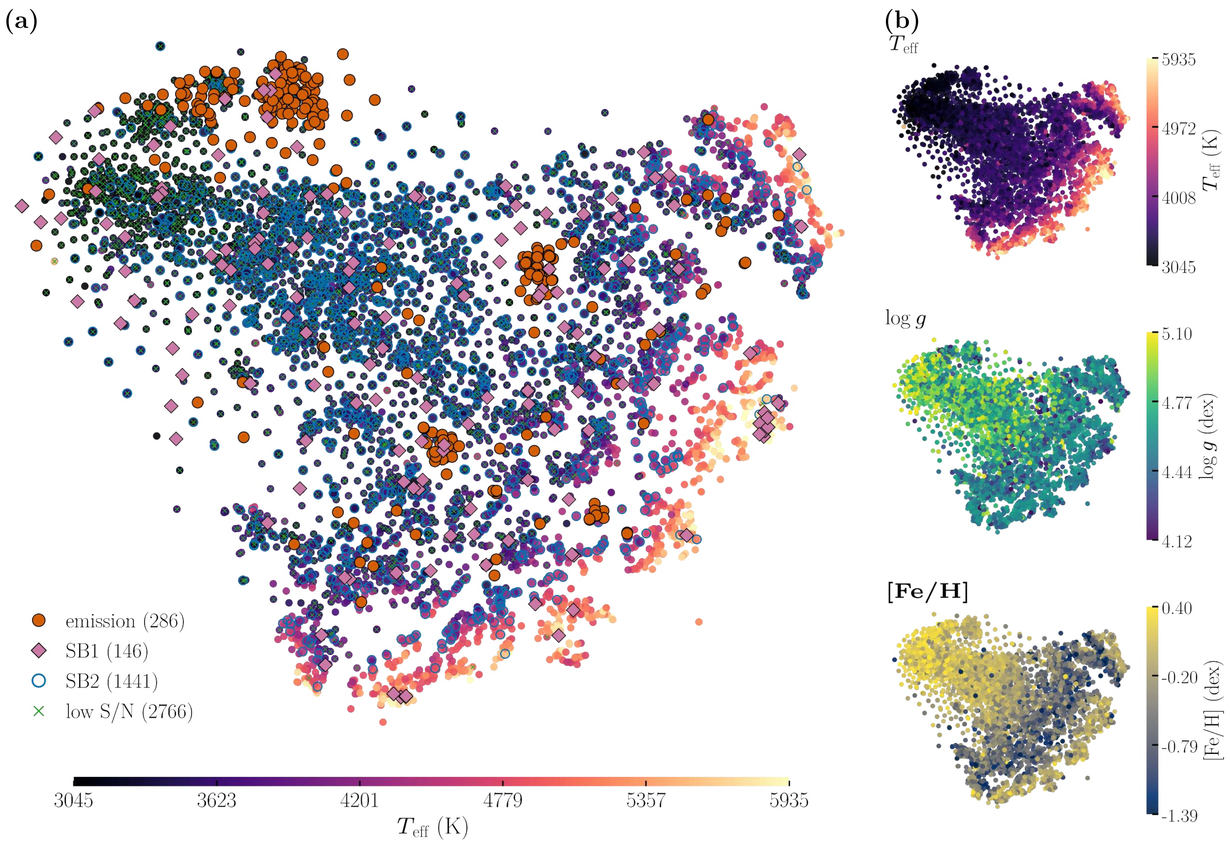


Figure 2 — Cuts and catalogue flags on the same map (grey: hidden stars). **(a)** Cuts: $T_{\rm eff} \geq 4500$ K, S/N ≥ 10 in 4 CCDs, and both; they hide stars, with no re-projection. **(b)** Selection by `flag_sp`: 0 = no warning; bit 1: Hα emission; 2: a CCD missing; 4, 8: SB1, SB2 binary candidates; 2048: low S/N (52% of the mapped stars, separate from our S/N cut). Flags are labels, not confirmed classes.

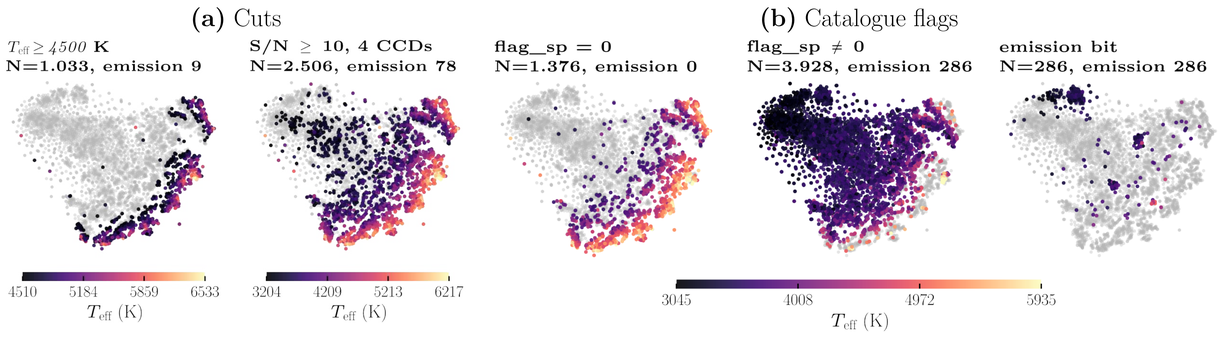


Figure 3 — Emission-flagged stars (286). **(a)** HDBSCAN finds five candidate subgroups: S0 (138 stars), S1 (11), S2 (54), S3 (14), S4 (29); 40 stay unassigned (dark grey; light grey: stars without the flag). **(b)** $T_{\rm eff}$: S0 concentrates cool M dwarfs (∼3400 K). **(c, d)** Kiel diagrams ($\log g$ vs $T_{\rm eff}$) over the 5 304 mapped stars (grey): S0 follows the solar-metallicity dwarf sequence, while S1 and S2 reach metal-poor regimes ($[Fe/H] < -0.5$). The subgroups are candidates; their nature is under investigation.

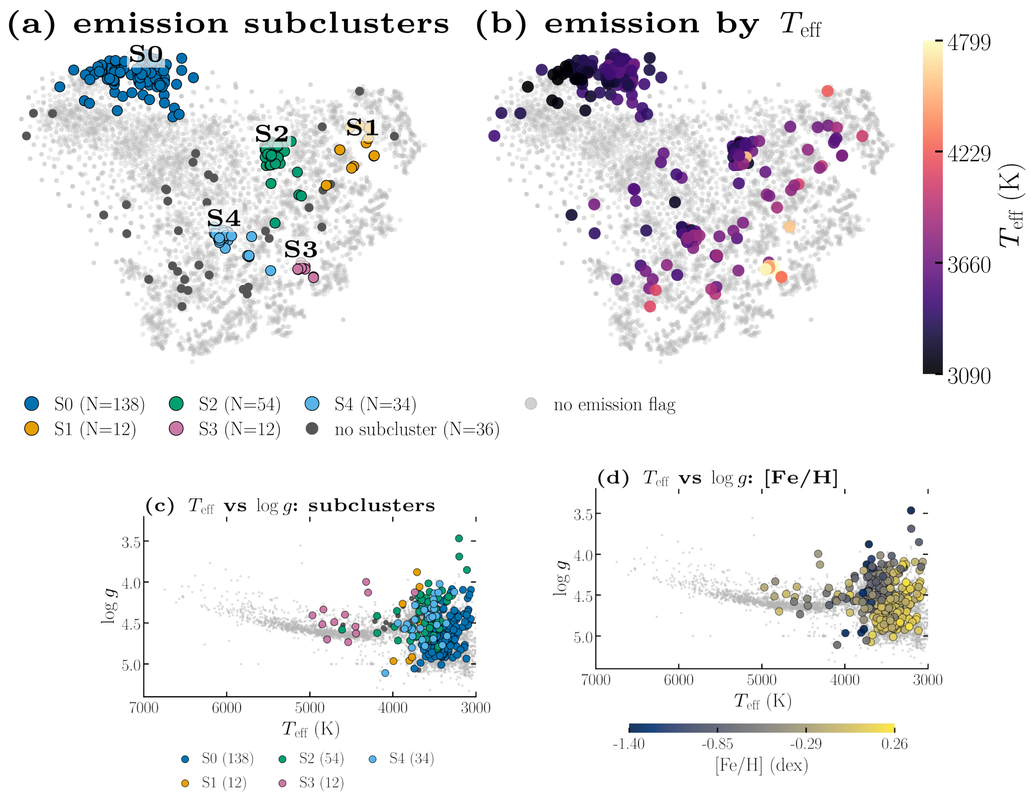


Figure 4 — Other settings: *t*-SNE (perplexity 30, 50) and UMAP (15, 30 neighbours) with a shared $T_{\rm eff}$ scale. Higher values raise the global distance correlation (up to 0.49); lower ones resolve finer substructure. Emission stars (orange) stay grouped in all four.

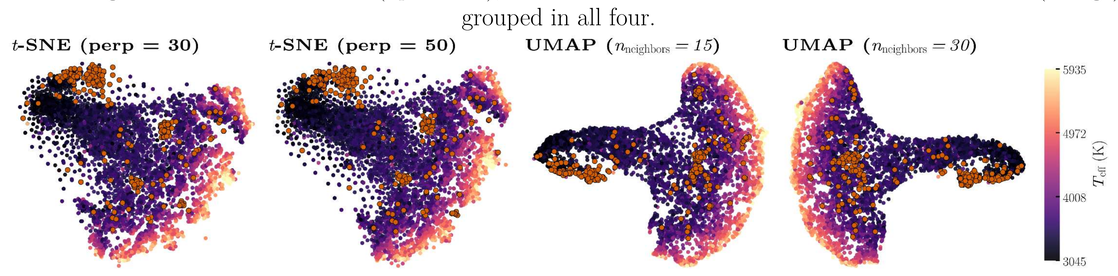
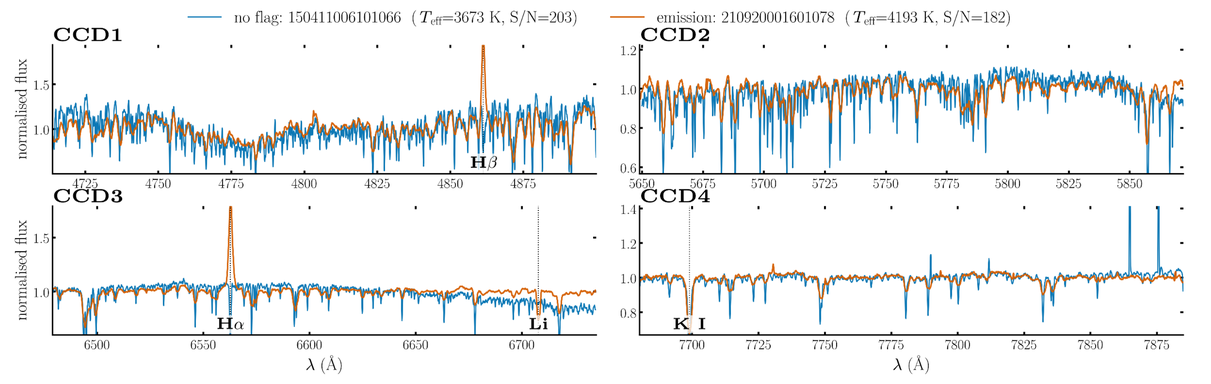


Figure 5 — Spectral verification across the four GALAH CCD channels (blue, green, red, and near-IR). Comparison between an emission candidate (210920001601078, $T_{\rm eff} = 4193$ K, S/N 182) and a quiet dwarf template (150411006101066, 3673 K, S/N 203). CCD3 shows pronounced Hα emission (6563 Å) and TiO band structure, supporting the catalogue flag for this candidate.



4. Key Findings

- **Unsupervised astrophysical recovery:** *t*-SNE and UMAP recover continuous $T_{\rm eff}$, $\log g$, and [Fe/H] gradients directly from 14 245-pixel spectra without stellar label supervision ($R^2 \approx 0.9$ for $T_{\rm eff}$).
- **Topology vs. locality trade-offs:** Perplexity 50 raises global distance correlation (0.43 vs. 0.38) with local overlap 0.36; UMAP reaches 0.44–0.49 globally but keeps less local overlap (0.21–0.25).
- **Emission sub-populations:** The 286 emission stars split into five candidate HDBSCAN subclusters: S0 (138 stars) corresponds to cool solar-metallicity dwarfs, while S1 and S2 represent metal-poor candidates.
- **Physical validation:** Spectral comparisons confirm authentic Hα emission and molecular line variations, supporting the catalogue flag for the inspected candidates.
- **Robustness:** emission stars stay grouped in every setting (perplexity 30, 50; UMAP 15, 30), while local and global structure vary with the settings.
- **Survey scalability:** Extending this unsupervised framework to all GALAH DR4 and searching for outliers will extend the census of the solar neighbourhood.

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